

1. What are the three main layers of software architecture? Write one line about each.

- **Presentation Layer (User Interface):** This top layer handles what the user sees on their screen, manages user interactions, and collects user inputs.
- **Business Logic Layer (Application Layer):** This middle layer processes the core functional rules, performs calculations, and controls the data flow between the other layers.
- **Data Access Layer (Storage Layer):** This bottom layer manages the connection, retrieval, modification, and permanent storage of information within the database.

2 Define client and server in your own words with one real-life example.

- **Client:** A client is the device or software application (like your laptop, smartphone, or a web browser) that requests a specific service, resource, or piece of data.
- **Server:** A server is a powerful computer or system that securely stores data, waits for incoming requests from clients, processes them, and delivers the requested resource back.
- **Real-Life Example:** Think of it like ordering food at a restaurant. You (the customer) are the **client** placing an order for a meal, while the kitchen is the **server** that receives your order, prepares the food, and sends it back to your table.

3 Write 2 differences each between desktop apps, web apps, and mobile apps.

- **Desktop Applications:** * They are installed directly onto a computer's operating system (like Windows or macOS) and typically offer high performance because they can fully access local hardware resources.
 - They usually run offline without requiring a constant internet connection, but updating them requires downloading and installing new version packages manually or via an installer.
- **Web Applications:** * They run entirely inside a web browser (like Chrome or Safari) and do not require any local installation, making them accessible from any device with an internet connection.
 - They require a continuous internet connection to function, but updates are seamless and automatic for the user since changes are deployed directly to the web server.
- **Mobile Applications:** * They are downloaded from specific app stores (like Google Play Store or Apple App Store) and are uniquely optimized for smartphone hardware, utilizing built-in features like GPS, cameras, and push notifications.
 - They are designed specifically for varying mobile screen sizes and touch-based interfaces, often offering limited offline functionality paired with cloud syncing.

4 Name any 3 databases and mention whether each is SQL or NoSQL.

- **MySQL:** This is a traditional **SQL** (Relational) database that stores data in structured, fixed tables with rows and columns, making it excellent for managing complex relationships and transactions.

- **MongoDB:** This is a popular **NoSQL** (Non-Relational) database that stores data as flexible, JSON-like documents, making it highly scalable and perfect for handling unstructured or rapidly changing data.
- **PostgreSQL:** This is an advanced **SQL** (Relational) database known for its reliability, extensive feature set, and strong support for complex data queries and enterprise-level applications.